DATA 621: Homework 3 — Logistic Regression

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1 Executive Summary

This analysis uses binary logistic regression to predict whether a neighborhood's crime rate is above the median (`target = 1`) or not (`target = 0`). Three candidate models are compared on a held-out portion of the training data. The selected model is then refit on all training observations and used to generate probabilities and classifications for the evaluation data. The required classification threshold is 0.5. Because the evaluation data do not include known target values, evaluation accuracy cannot be calculated.

2 1. Data Exploration

2.1 1.1 Dataset size and variables

Table 1: Dataset size and missing cells

Dataset	Rows	Columns	Missing_Cells
Training	466	13	0
Evaluation	40	12	0

The training file contains 466 observations and 13 variables. The evaluation file contains 40 observations and 12 predictor variables. The response variable, `target`, indicates whether crime is above the median. Predictors are `zn` (large-lot residential zoning), `indus` (non-retail business land), `chas` (Charles River adjacency), `nox` (nitrogen oxides), `rm` (average rooms), `age` (older owner-occupied housing), `dis` (distance to employment centers), `rad` (highway access), `tax` (property-tax rate), `ptratio` (pupil-teacher ratio), `lstat` (lower-status population percentage), and `medv` (median home value).

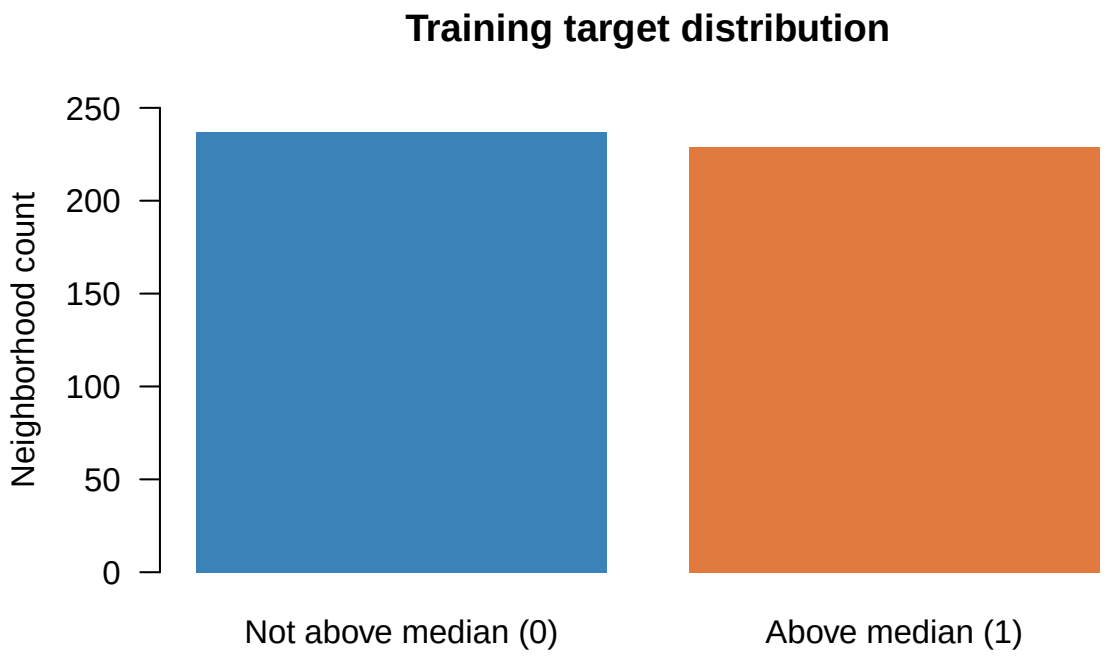
2.2 1.2 Summary statistics and target balance

Table 2: Training predictor summary statistics

	Variable	Mean	SD	Median	Min	Max
zn	zn	11.577	23.365	0.000	0.000	100.000
indus	indus	11.105	6.846	9.690	0.460	27.740
chas	chas	0.071	0.257	0.000	0.000	1.000
nox	nox	0.554	0.117	0.538	0.389	0.871
rm	rm	6.291	0.705	6.210	3.863	8.780
age	age	68.368	28.321	77.150	2.900	100.000
dis	dis	3.796	2.107	3.191	1.130	12.126
rad	rad	9.530	8.686	5.000	1.000	24.000
tax	tax	409.502	167.900	334.500	187.000	711.000
ptratio	ptratio	18.398	2.197	18.900	12.600	22.000
lstat	lstat	12.631	7.102	11.350	1.730	37.970
medv	medv	22.589	9.240	21.200	5.000	50.000

Table 3: Target distribution

Target	Count	Percent
0: Not above median	237	50.9
1: Above median	229	49.1



The target balance matters because accuracy can be misleading when one class is more common. Sensitivity, specificity, precision, F1 score, and AUC are also used to compare models.

2.3 1.3 Predictor distributions and correlations

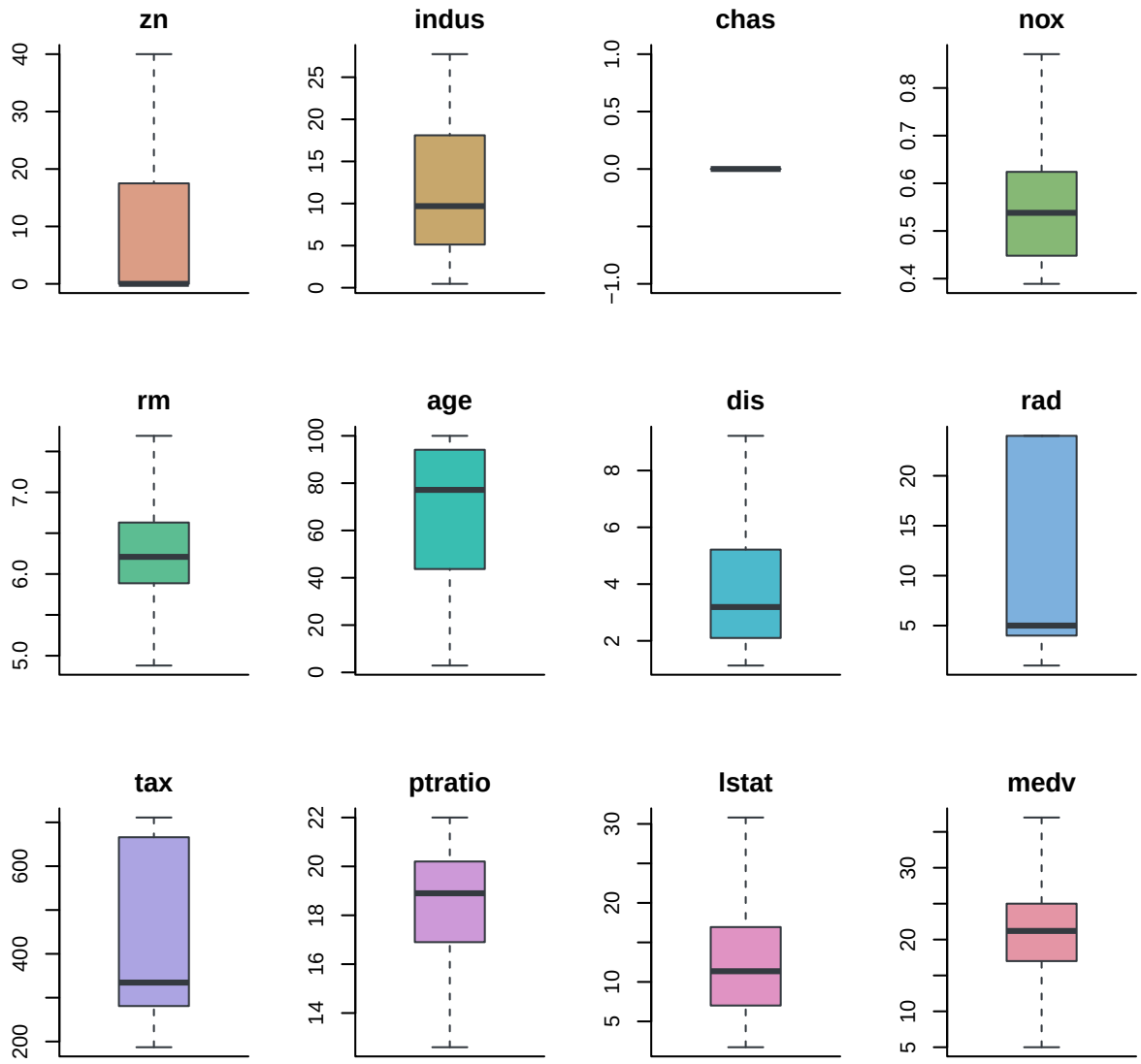
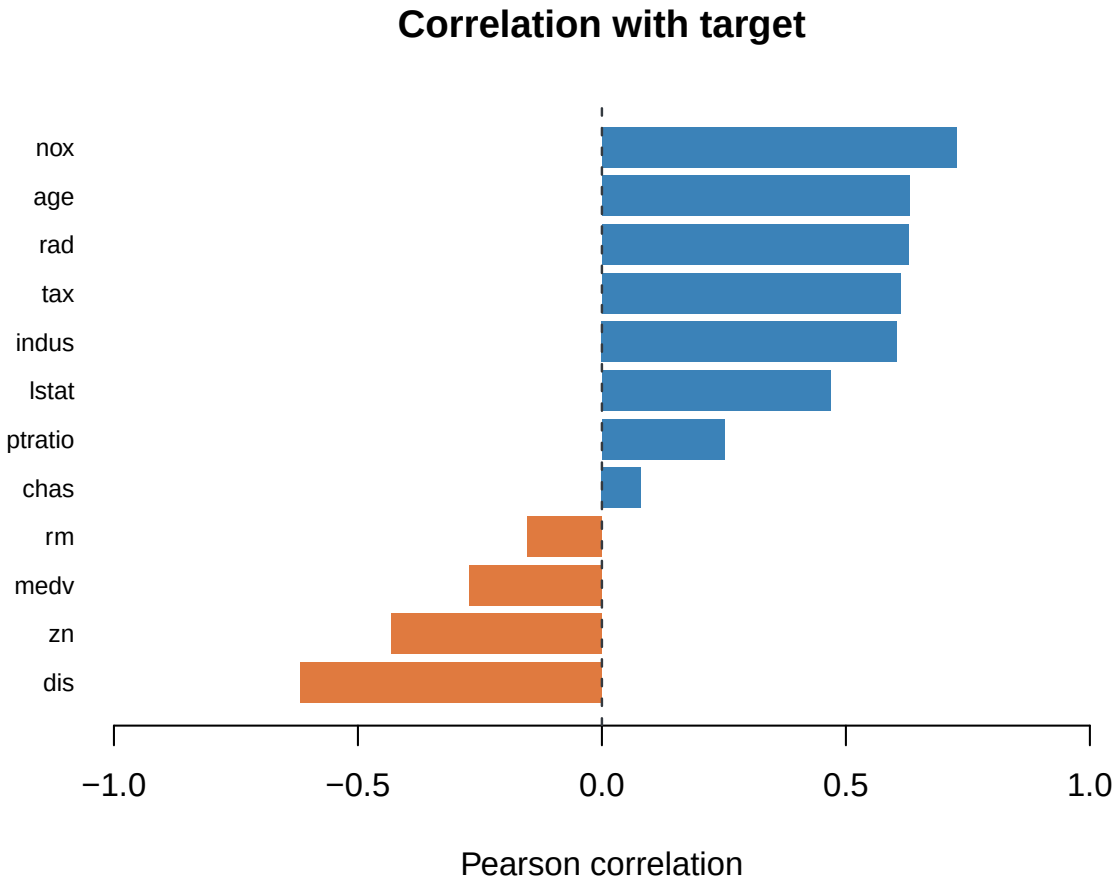


Table 4: Pearson correlations among predictors

	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	lstat	medv
zn	1.00	-0.54	-0.04	-0.52	0.32	-0.57	0.66	-0.32	-0.32	-0.39	-0.43	0.38
indus	-0.54	1.00	0.06	0.76	-0.39	0.64	-0.70	0.60	0.73	0.39	0.61	-0.50
chas	-0.04	0.06	1.00	0.10	0.09	0.08	-0.10	-0.02	-0.05	-0.13	-0.05	0.16
nox	-0.52	0.76	0.10	1.00	-0.30	0.74	-0.77	0.60	0.65	0.18	0.60	-0.43
rm	0.32	-0.39	0.09	-0.30	1.00	-0.23	0.20	-0.21	-0.30	-0.36	-0.63	0.71
age	-0.57	0.64	0.08	0.74	-0.23	1.00	-0.75	0.46	0.51	0.26	0.61	-0.38
dis	0.66	-0.70	-0.10	-0.77	0.20	-0.75	1.00	-0.49	-0.53	-0.23	-0.51	0.26
rad	-0.32	0.60	-0.02	0.60	-0.21	0.46	-0.49	1.00	0.91	0.47	0.50	-0.40
tax	-0.32	0.73	-0.05	0.65	-0.30	0.51	-0.53	0.91	1.00	0.47	0.56	-0.49
ptratio	-0.39	0.39	-0.13	0.18	-0.36	0.26	-0.23	0.47	0.47	1.00	0.38	-0.52
lstat	-0.43	0.61	-0.05	0.60	-0.63	0.61	-0.51	0.50	0.56	0.38	1.00	-0.74
medv	0.38	-0.50	0.16	-0.43	0.71	-0.38	0.26	-0.40	-0.49	-0.52	-0.74	1.00

Table 5: Predictor correlations with target

Variable	Correlation_with_target
dis	-0.619
zn	-0.432
medv	-0.271
rm	-0.153
chas	0.080
ptratio	0.251
lstat	0.469
indus	0.605
tax	0.611
rad	0.628
age	0.630
nox	0.726



The boxplots show that the predictors do not all have similarly shaped distributions. In particular, `zn` is strongly right-skewed: many observations are at or near zero, while a smaller number have much larger values. `rad` is also unevenly distributed and takes a limited set of discrete values, with observations concentrated at particular highway-access levels. These patterns matter because extreme values and skewness can influence the scale of a predictor and make a one-unit change less comparable across the range of the data. The boxplots are therefore useful not only for identifying unusual values, but also for considering whether transformations might make some distributions less skewed.

A logarithmic transformation is a reasonable sensitivity check for nonnegative, right-skewed variables such as `zn` and `rad`. Because `zn` includes zero values, `log1p(zn)` (the natural logarithm of one plus the predictor) can be used without dropping those observations. The same transformation is applied to `rad` for a consistent comparison. The original predictors are retained in the primary models so that their coefficients remain expressed on the supplied measurement scales; the transformed version is assessed separately below.

Correlations with the binary target are useful for exploration, but they do not establish causation and do not account for relationships among predictors. Correlated predictors can also make individual regression coefficients less stable.

3 2. Data Preparation

The two files contain the same 12 predictors; only the training data include `target`. If missing values are present, predictor missingness is filled using training-set medians, which are also applied to the evaluation set. A missing response, if any, is filled with the most frequent observed target class. The supplied data are otherwise kept in their original units to preserve interpretability. The `chas` indicator remains coded as 0/1.

A stratified split reserves approximately 20% of the training data for validation. Stratification preserves the target-class proportions approximately, and all models are evaluated on the same held-out observations.

Table 6: Stratified modeling and validation split

Subset	Rows	Target_0	Target_1
Model fitting	374	190	184
Validation	92	47	45

4 3. Build Models

Three primary candidates are compared: (1) a full model with all 12 predictors; (2) a focused model using `nox`, `rm`, `dis`, `rad`, `tax`, `ptratio`, `lstat`, and `medv`; and (3) a stepwise model selected from the full model using Akaike's Information Criterion (AIC). The focused model was designed to retain predictors that represent several substantively relevant dimensions of neighborhood conditions: air quality (`nox`), housing characteristics (`rm` and `medv`), access and location (`dis` and `rad`), and socioeconomic or public-service conditions (`tax`, `ptratio`, and `lstat`). This provides a more compact model than the full specification while preserving a range of environmental, geographic, housing, and socioeconomic information. The four omitted variables (`zn`, `indus`, `chas`, and `age`) are left out to test whether a narrower, conceptually grouped set of predictors can provide useful predictive performance. This is a modeling choice rather than a claim that the omitted variables are unimportant.

A separate transformed full model is also fitted as a sensitivity check. It replaces `zn` and `rad` with `log1p(zn)` and `log1p(rad)`, respectively. This comparison helps assess whether reducing the influence of their right-skewed scales changes validation performance. It is not added to the three primary candidates used for the assignment's model-selection comparison. AIC balances model fit with model complexity.

For logistic regression, a coefficient is the change in log-odds for a one-unit predictor increase while holding other included predictors constant. Exponentiating a coefficient gives its odds ratio. Positive coefficients indicate higher modeled odds of `target = 1`; negative coefficients indicate lower modeled odds. These are associations, not causal effects.

```
**Model 1: Full coefficients and odds ratios**
```

Term	Estimate	Std_Error	z_value	p_value	Odds_Ratio
(Intercept)	-38.6070	7.2513	-5.3242	0.0000	0.0000
zn	-0.0656	0.0390	-1.6815	0.0927	0.9365
indus	-0.0806	0.0540	-1.4919	0.1357	0.9226
chas	1.4474	0.8548	1.6933	0.0904	4.2522
nox	48.9279	8.7310	5.6039	0.0000	177471420872358153424 8.0000
rm	-0.8502	0.8090	-1.0509	0.2933	0.4273
age	0.0279	0.0148	1.8827	0.0597	1.0283
dis	0.6267	0.2589	2.4207	0.0155	1.8715
rad	0.6442	0.1926	3.3448	0.0008	1.9044
tax	-0.0068	0.0034	-1.9833	0.0473	0.9933
ptratio	0.4244	0.1432	2.9641	0.0030	1.5287
lstat	0.0623	0.0611	1.0203	0.3076	1.0643
medv	0.1852	0.0730	2.5377	0.0112	1.2035

Model 2: Focused coefficients and odds ratios

Term	Estimate	Std_Error	z_value	p_value	Odds_Ratio
(Intercept)	-33.8452	6.4416	-5.2542	0.0000	0.0000
nox	43.0907	7.2035	5.9819	0.0000	5176480055973100544.0000
rm	-0.2746	0.6629	-0.4143	0.6787	0.7599
dis	0.2517	0.1963	1.2826	0.1996	1.2862
rad	0.6971	0.1652	4.2198	0.0000	2.0080
tax	-0.0091	0.0028	-3.2621	0.0011	0.9909
ptratio	0.3562	0.1259	2.8293	0.0047	1.4280
lstat	0.1029	0.0502	2.0507	0.0403	1.1083
medv	0.1238	0.0608	2.0382	0.0415	1.1318

Model 3: Stepwise AIC coefficients and odds ratios

Term	Estimate	Std_Error	z_value	p_value	Odds_Ratio
(Intercept)	-36.4431	6.8669	-5.3070	0.0000	0.0000
zn	-0.0612	0.0380	-1.6100	0.1074	0.9406
indus	-0.0778	0.0537	-1.4499	0.1471	0.9251
chas	1.5326	0.8739	1.7539	0.0795	4.6303
nox	48.9096	8.7323	5.6010	0.0000	174242871821520679738 2.0000
rm	-1.1831	0.7378	-1.6036	0.1088	0.3063
age	0.0348	0.0133	2.6065	0.0091	1.0354
dis	0.6335	0.2595	2.4407	0.0147	1.8841
rad	0.6480	0.1905	3.4022	0.0007	1.9117
tax	-0.0065	0.0034	-1.9252	0.0542	0.9935
ptratio	0.4277	0.1434	2.9820	0.0029	1.5337
medv	0.1838	0.0733	2.5070	0.0122	1.2018

In interpreting the coefficient tables, it is important to distinguish a predictor's simple correlation with the target from its logistic-regression coefficient. A correlation describes the unadjusted relationship between one predictor and the target. A regression coefficient describes the predictor's association with the target after holding the other

included predictors constant. When predictors are correlated with one another, this adjustment can change the size of a coefficient and can even reverse its sign. This is a possible explanation for the differences observed for `dis`, `tax`, and `medv`: their coefficients represent conditional relationships in a model containing other related neighborhood and housing measures, rather than their separate, unadjusted relationships with the target. Such sign reversals can arise from shared information among predictors or suppression effects; they do not necessarily indicate an error in the model. The coefficients should therefore be interpreted together with the correlation matrix and with caution about multicollinearity.

Statistical significance is not the only selection criterion. Predictive performance on held-out observations, model parsimony, and whether estimated relationships are plausible are also considered.

5 4. Select and Evaluate Models

All classification metrics use a 0.5 threshold, with `target = 1` as the positive class. Accuracy is the proportion correctly classified; classification error is the proportion misclassified; precision is the proportion of predicted positives that are true positives; sensitivity is the proportion of actual positives identified; specificity is the proportion of actual negatives identified; and F1 is the harmonic mean of precision and sensitivity. AUC measures ranking performance across thresholds. Lower AIC is preferred when comparing models fitted to the same data.

Table 7: Validation performance at the 0.5 threshold

Model	AIC	TP	TN	FP	FN	Accuracy	Error	Precision	Sensitivity	Specificity	F1	AUC
Model 1: Full	175.213	38	44	3	7	0.891	0.109	0.927	0.844	0.936	0.884	0.960
Model 2: Focused	181.656	35	43	4	10	0.848	0.152	0.897	0.778	0.915	0.833	0.960
Model 3: Stepwise AIC	174.238	40	44	3	5	0.913	0.087	0.930	0.889	0.936	0.909	0.963

Model 1: Full confusion matrix

	0	1
0	44	3
1	7	38

Model 2: Focused confusion matrix

	0	1
0	43	4
1	10	35

Model 3: Stepwise AIC confusion matrix

	0	1
0	44	3
1	5	40

The transformed-model comparison below is a sensitivity check, not a fourth primary candidate. It uses the same validation observations and the same 0.5 threshold as the original full model. If the transformed model performs better, that suggests the transformation may be useful for prediction; if performance is similar or worse, retaining the original scales is reasonable and keeps the coefficients easier to interpret. The transformation comparison should be considered alongside the distribution plots rather than treated as an automatic reason to transform every skewed predictor.

Table 8: Original and log-transformed full-model sensitivity comparison

Model	AIC	TP	TN	FP	FN	Accuracy	Error	Precision	Sensitivity	Specificity	F1	AUC
Original full model	175.213	38	44	3	7	0.891	0.109	0.927	0.844	0.936	0.884	0.960
Transformed full model (log1p zn, log1p rad)	174.635	40	44	3	5	0.913	0.087	0.930	0.889	0.936	0.909	0.962

The preferred model is selected using validation AUC first, then F1 score, then accuracy. This emphasizes both ranking quality and classification performance rather than relying on accuracy alone. AIC is supporting evidence; it is not directly used as the only criterion because the candidate models can have different numbers of predictors.

```
Selected model based on validation ranking: Model 3: Stepwise AIC
```

6 5. Evaluation-Set Predictions

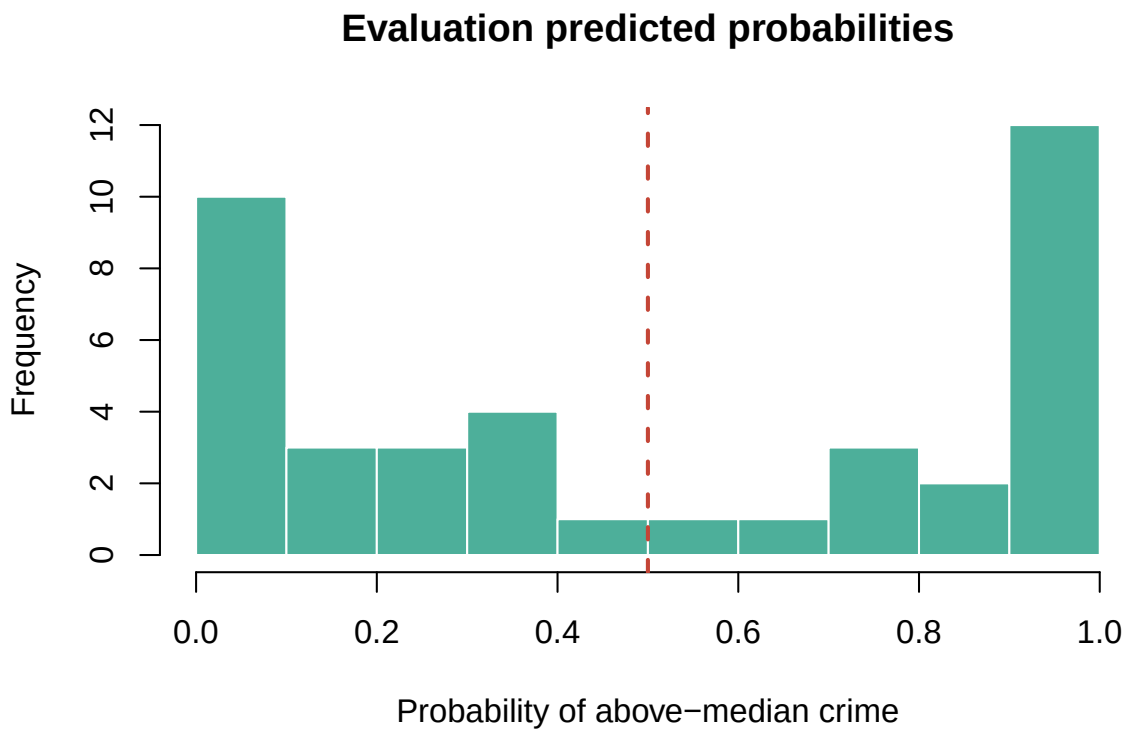
The selected model specification is refit on all training observations. The final model generates probabilities for the evaluation observations, and the required threshold of 0.5 converts probabilities into class labels.

Table 9: First 15 evaluation predictions; complete output saved to CSV

Evaluation_Row	Probability_Above_Median_Crime	Predicted_Class_0_5
1	0.0397	0
2	0.7023	1
3	0.7209	1
4	0.4049	0
5	0.0980	0
6	0.3359	0
7	0.3549	0
8	0.0091	0
9	0.0038	0
10	0.0014	0
11	0.2872	0
12	0.2321	0
13	0.8507	1
14	0.6988	1
15	0.5496	1

Table 10: Evaluation predictions by class

Predicted_Class	Count	Percent
0	21	52.5
1	19	47.5



The full 40-row output is written to `crime-evaluation-predictions.csv` when the document is knitted. The probability column preserves the estimated probability, while the class column applies the required threshold. These are predictions, not verified outcomes; the evaluation file contains no known target values with which to calculate accuracy.

7 Conclusion

This analysis used binary logistic regression to classify neighborhoods according to whether their crime rate was above the median. The exploratory analysis summarized the predictor variables, examined the balance of the two target classes, and assessed missing values, distributions, and correlations. These steps helped identify the structure and quality of the data before fitting the models. The predictors were retained in their original units, and a stratified validation sample was used so that the candidate models could be compared on the same held-out observations.

Three models were evaluated: a full model containing all 12 predictors, a focused model containing eight selected predictors, and a stepwise model selected using AIC. At the required 0.5 classification threshold, the stepwise AIC model performed best on the validation sample. It achieved 91.3% accuracy, compared with 89.1% for the full model and 84.8% for the focused model. The stepwise model also had the highest sensitivity (88.9%), F1 score (0.909), and AUC (0.963). Its confusion matrix showed 44 true negatives, 40 true positives, three false positives, and five false negatives. In practical terms, it correctly classified 84 of the 92 validation observations and missed fewer above-median cases than either alternative.

These results support selecting the stepwise model from the three candidates for scoring the evaluation dataset. However, the differences in validation performance should be interpreted with care: the validation set is a subset of the supplied training data, and results on new data may differ. The model identifies statistical associations in the supplied variables; it does not show that any predictor causes crime levels to rise or fall. In addition, the 0.5 threshold is the threshold required for this assignment, but a real-world application might choose a different threshold depending on the relative costs of false positives and false negatives.

Finally, the selected model was refit using all available training observations and used to produce a probability and a class prediction for each evaluation record. The full predictions are saved to `crime-evaluation-predictions.csv`. Because the evaluation dataset does not contain known target values, its accuracy, sensitivity, specificity, and other outcome-based metrics cannot be calculated. The predictions should therefore be treated as model estimates rather than confirmed classifications.

8 Appendix A. R Code

The appendix provides the R code in a small, line-wrapping listing. The PDF `listings` setting allows long lines to wrap and code listings to continue across pages. The code is displayed for reference and is not re-executed in the appendix.

```
# DATA 621 Homework 3: Logistic Regression
# CSV files are read from the Downloads folder paths below.

knitr::opts_chunk$set(echo=FALSE,warning=FALSE,message=FALSE,
  comment="",fig.width=6.2,fig.height=4.2,fig.align="center",
  out.width="0.92\\linewidth")
options(width=75,scipen=999)
set.seed(621)

train_raw <- read.csv("C:/Users/Home/Downloads/crime-training-data_modified.csv",
  stringsAsFactors=FALSE)
eval_raw <- read.csv("C:/Users/Home/Downloads/crime-evaluation-data_modified.csv",
  stringsAsFactors=FALSE)
predictors <- setdiff(names(train_raw),"target")
train_raw$target <- as.integer(train_raw$target)

# Explore missingness, summary statistics, target balance, plots,
# and predictor correlations as shown in Sections 1 and 2 above.

if (anyNA(train_raw)||anyNA(eval_raw)) {
  for (v in predictors) {
    med <- median(train_raw[[v]],na.rm=TRUE)
    train_raw[[v]][is.na(train_raw[[v]])] <- med
    eval_raw[[v]][is.na(eval_raw[[v]])] <- med
  }
  if (anyNA(train_raw$target)) {
    train_raw$target[is.na(train_raw$target)] <- as.integer(
      names(which.max(table(train_raw$target))))
  }
}

# Stratified 80/20 validation split
set.seed(621)
class0 <- which(train_raw$target==0)
class1 <- which(train_raw$target==1)
valid_idx <- sort(c(sample(class0,max(1,floor(.20*length(class0)))),
                    sample(class1,max(1,floor(.20*length(class1))))))
model_idx <- setdiff(seq_len(nrow(train_raw)),valid_idx)
model_data <- train_raw[model_idx,,drop=FALSE]
valid_data <- train_raw[valid_idx,,drop=FALSE]

# Model 1: full predictor model
model_full <- glm(target~zn+indus+chas+nox+rm+age+dis+rad+tax+
```

```
  ptratio+lstat+medv,data=model_data,family=binomial(link="logit"))

# Model 2: focused predictor model
model_focused <- glm(target~nox+rm+dis+rad+tax+ptratio+lstat+medv,
  data=model_data,family=binomial(link="logit"))

# Model 3: stepwise AIC model
model_stepwise <- step(model_full,direction="both",trace=0)

# Sensitivity model: log-transform the two predictors with notably
# uneven/right-skewed distributions while retaining the other predictors.
model_data_transformed <- model_data
model_data_transformed$zn_log1p <- log1p(model_data_transformed$zn)
model_data_transformed$rad_log1p <- log1p(model_data_transformed$rad)
model_transformed_full <- glm(
  target~zn_log1p+indus+chas+nox+rm+age+dis+rad_log1p+tax+
    ptratio+lstat+medv,
  data=model_data_transformed,family=binomial(link="logit"))

models <- list("Model 1: Full"=model_full,
  "Model 2: Focused"=model_focused,"Model 3: Stepwise AIC"=model_stepwise)

safe_div <- function(a,b) if (b==0) NA_real_ else a/b
auc_rank <- function(actual,prob) {
  n1 <- sum(actual==1); n0 <- sum(actual==0)
  if (n1==0||n0==0) return(NA_real_)
  r <- rank(prob,ties.method="average")
  (sum(r[actual==1])-n1*(n1+1)/2)/(n1*n0)
}
get_metrics <- function(actual,prob,threshold=0.5) {
  pred <- as.integer(prob>=threshold)
  TP <- sum(pred==1 & actual==1); TN <- sum(pred==0 & actual==0)
  FP <- sum(pred==1 & actual==0); FN <- sum(pred==0 & actual==1)
  precision <- safe_div(TP,TP+FP)
  sensitivity <- safe_div(TP,TP+FN)
  specificity <- safe_div(TN,TN+FP)
  f1 <- if (is.na(precision)||is.na(sensitivity)||precision+sensitivity==0)
    NA_real_ else 2*precision*sensitivity/(precision+sensitivity)
  data.frame(TP=TP,TN=TN,FP=FP,FN=FN,
    Accuracy=(TP+TN)/length(actual),
    Classification_Error=(FP+FN)/length(actual),
    Precision=precision,Sensitivity=sensitivity,Specificity=specificity,
    F1=f1,AUC=auc_rank(actual,prob))
}
validation_metrics <- do.call(rbind,lapply(names(models),function(nm) {
  fit <- models[[nm]]
  p <- predict(fit,newdata=valid_data,type="response")
  data.frame(Model=nm,AIC=AIC(fit),
    get_metrics(valid_data$target,p,threshold=0.5),
    row.names=NULL,check.names=FALSE)
}))
rownames(validation_metrics) <- NULL

# Confusion matrices and model ranking
for (nm in names(models)) {
  p <- predict(models[[nm]],newdata=valid_data,type="response")
```

```
  predicted <- factor(ifelse(p>=0.5,1,0),levels=c(0,1))
  actual <- factor(valid_data$target,levels=c(0,1))
  print(table(Actual=actual,Predicted=predicted))
}
rank_value <- function(x) ifelse(is.na(x),-Inf,x)
ord <- order(rank_value(validation_metrics$AUC),
  rank_value(validation_metrics$F1),rank_value(validation_metrics$Accuracy),
  decreasing=TRUE)
selected_name <- validation_metrics$Model[ord[1]]

# Refit selected specification and score evaluation rows
selected_formula <- formula(models[[selected_name]])
final_model <- glm(selected_formula,data=train_raw,
  family=binomial(link="logit"))
eval_prob <- predict(final_model,newdata=eval_raw,type="response")
eval_class <- ifelse(eval_prob>=0.5,1L,0L)
evaluation_predictions <- data.frame(
  Evaluation_Row=seq_len(nrow(eval_raw)),
  Probability_Above_Median_Crime=eval_prob,
  Predicted_Class_0_5=eval_class)
write.csv(evaluation_predictions,"crime-evaluation-predictions.csv",
  row.names=FALSE)
```